

LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO: 39. Vehicle Detection Using OpenCV

Aim

To detect moving vehicles in a video using **OpenCV**.

Algorithm

1. Import OpenCV.
2. Open the car video **VEHICLE.mp4**.
3. Read each video frame.
4. Apply background subtraction.
5. Remove small noise using morphological operations.
6. Find contours of moving objects.
7. Draw rectangles around large moving objects.
8. Display the detected vehicles.
9. Press **ESC** to stop.

CODE:

```
EXP-39.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...
File Edit Format Run Options Window Help

# Detect vehicles
for contour in contours:

    area = cv2.contourArea(contour)

    if area > 500:

        x, y, w, h = cv2.boundingRect(contour)

        # Draw rectangle
        cv2.rectangle(
            frame,
            (x, y),
            (x + w, y + h),
            (0, 255, 0),
            2
        )

        # Display label
        cv2.putText(
            frame,
            "Vehicle",
            (x, y - 10),
            cv2.FONT_HERSHEY_SIMPLEX,
            0.6,
            (0, 255, 0),
            2
        )

# Show result
cv2.imshow("Vehicle Detection", frame)

# Press ESC to exit
if cv2.waitKey(30) & 0xFF == 27:
    break

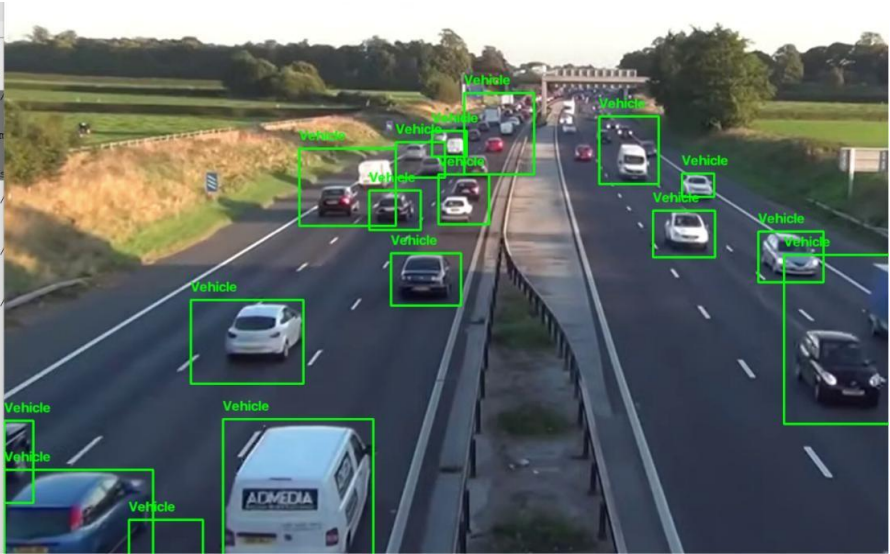
cap.release()
cv2.destroyAllWindows()

print("Vehicle detection completed successfully!")
```

Ln: 70 Col: 0

OUTPUT :

```
"IDLE Shell 3.13.3"
File Edit Shell Debug Options Window Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025,
AMD64) on win32
Enter "help" below or click "Help" above for more
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/
periments/EXP-38.py
Traceback (most recent call last):
  File "C:/Users/POORNA SKHT/OneDrive/Tài liệu/comp
iments/EXP-38.py", line 15, in <module>
    face_cascade = cv2.CascadeClassifier(
AttributeError: module 'cv2' has no attribute 'Cas
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/
periments/EXP-38.py
Face/object detected successfully!
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/
periments/EXP-39.py
Vehicle detection completed successfully!
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/
periments/EXP-39.py
```



Result

The moving vehicles in VEHICLE.mp4 are successfully detected and marked with rectangular bounding boxes using OpenCV.